FARM MANAGEMENT SYSTEM

Salesforce-Based Agricultural Data Management Project

Farm Management System ■ Alternative Version

Salesforce-Based Agricultural Data Management

Project

Salesforce-Based Agricultural Data Management Project

The Farm Management System is a Salesforce-based project created to organize agricultural information in a centralized digital environment. The project focuses on Farm, Crop, Farmer and Buyer records and connects them through a structured

data model. It supports organized data entry, record updates, relationships, access control, reports and dashboards. The documentation covers the complete project journey from problem identification and requirements to Salesforce configuration, security, testing and future scope. The purpose of this alternative version is to present the same project in a detailed academic format with fuller written explanations. Each page contains expanded project content so that the documentation can be used for project submission, review and presentation. The project demonstrates how Salesforce features can be combined to manage agricultural information systematically.

PROJECT DETAILS

■ The Farm Management System is a Salesforce-based project created to organize agricultural information in a centralized

digital environment.

■ The project focuses on Farm, Crop, Farmer and Buyer records and connects them through a structured data model.

■ It supports organized data entry, record updates, relationships, access control, reports and dashboards.

■ The documentation covers the complete project journey from problem identification and requirements to Salesforce configuration, security, testing and future scope.

Section relevance: The information in Salesforce-Based Agricultural Data Management Project forms part of the Farm Management System documentation and is connected to the project's overall data management, Salesforce configuration, security, reporting, testing or future-scope structure as described in this report. This section is intended to make the project easier to review, present and understand.

Review note: The project terminology and scope are retained consistently across the documentation. The section can be used as part of the final project explanation and demonstration.

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Project User Story

In the Farm Management System, a farmer, farm manager, buyer or system administrator needs a centralized Salesforce-based platform to record and manage agricultural information. The user should be able to maintain farm details, crop production records, farmer information and buyer details in a systematic way. The application should make it easier to enter new information and update existing records without depending on separate manual documents.

Proper field configuration and validation should help prevent incorrect or inconsistent entries. The system should connect related information so that users can understand which farmer manages a farm and which crops belong to a farm. Reports should provide detailed views of stored information, while dashboards should provide quick summaries.

Role-based access should help protect records and support collaboration. The overall goal is to reduce manual work, improve organization and support data-driven farm management.

PROJECT DETAILS

■ In the Farm Management System, a farmer, farm manager, buyer or system administrator needs a centralized Salesforce-based platform to record and manage agricultural information.

■ The user should be able to maintain farm details, crop production records, farmer information and buyer details in a systematic way.

■ The application should make it easier to enter new information and update existing records without depending on separate

manual documents.

■ Proper field configuration and validation should help prevent incorrect or inconsistent entries.

Section relevance: The information in Project User Story forms part of the Farm Management System documentation and is connected to the project's overall data management, Salesforce configuration, security, reporting, testing or future-scope structure as described in this report. This section is intended to make the project easier to review, present and understand.

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Executive Summary

The Farm Management System brings important agricultural records into a single Salesforce environment. The project is organized around four main custom objects: Farm, Crop, Farmer and Buyer. These objects provide a structured foundation for storing information and connecting related records.

Custom tabs and a Lightning App provide navigation so users can reach the main information areas easily. Fields and relationships define the data model, while profiles, roles, users and sharing rules support controlled access. Reports organize detailed information and dashboards provide summarized views.

Testing is included to check objects, fields, relationships, permissions and analytics. The project documentation also covers

user adoption, advantages, limitations, conclusion and future scope. Together, these components describe a complete Salesforce-based approach to agricultural information management and provide a clear structure for academic project documentation.

PROJECT DETAILS

■ The Farm Management System brings important agricultural records into a single Salesforce environment.

■ The project is organized around four main custom objects: Farm, Crop, Farmer and Buyer.

■ These objects provide a structured foundation for storing information and connecting related records.

■ Custom tabs and a Lightning App provide navigation so users can reach the main information areas easily.

Section relevance: The information in Executive Summary forms part of the Farm Management System documentation and is connected to the project's overall data management, Salesforce configuration, security, reporting, testing or future-scope structure as described in this report. This section is intended to make the project easier to review, present and understand.

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1. Introduction

Agriculture produces information related to farms, crops, farmers, buyers and production activities. When this information is maintained through separate notebooks, files or disconnected records, users may spend additional time searching for information and updating multiple places. The Farm Management System is designed to provide a structured digital workspace for this information using Salesforce.

The project brings farm, crop, farmer and buyer records into one centralized environment. This allows users to work with related information instead of maintaining isolated records. Salesforce custom objects provide the main data structure, while fields capture the details required for each record.

Relationships connect information across the objects and make navigation easier. Security features help control access according to responsibilities. Reports and dashboards provide organized information for review and analysis. The introduction therefore establishes the foundation for a systematic method of managing farm-related data.

PROJECT DETAILS

■ Agriculture produces information related to farms, crops, farmers, buyers and production activities.

■ When this information is maintained through separate notebooks, files or disconnected records, users may spend additional

time searching for information and updating multiple places.

■ The Farm Management System is designed to provide a structured digital workspace for this information using Salesforce.

■ The project brings farm, crop, farmer and buyer records into one centralized environment.

Section relevance: The information in 1. Introduction forms part of the Farm Management System documentation and is connected to the project's overall data management, Salesforce configuration, security, reporting, testing or future-scope



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2. Project Purpose

The purpose of the Farm Management System is to make agricultural information easier to enter, update, search and analyze through Salesforce. The project aims to replace scattered manual handling with a structured digital process. Farm details are stored in the Farm object, production information is maintained through the Crop object, farmer details are recorded through the Farmer object and buyer information is maintained through the Buyer object.

The purpose also includes creating relationships between records so information can be reviewed together. Another purpose is to provide suitable access for different users through Salesforce security features. Reports help users inspect detailed information, while dashboards provide summarized views for monitoring.

The system is also intended to improve consistency by using defined field types and controlled values. By bringing these functions together, the project creates a practical foundation for organized agricultural information management.

PROJECT DETAILS

■ The purpose of the Farm Management System is to make agricultural information easier to enter, update, search and analyze through Salesforce.

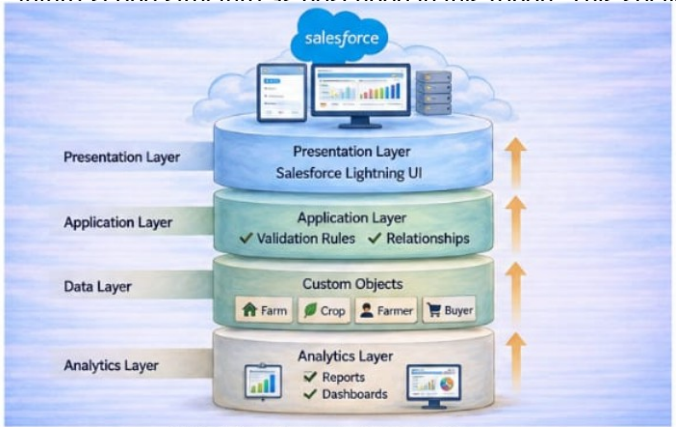
■ The project aims to replace scattered manual handling with a structured digital process.

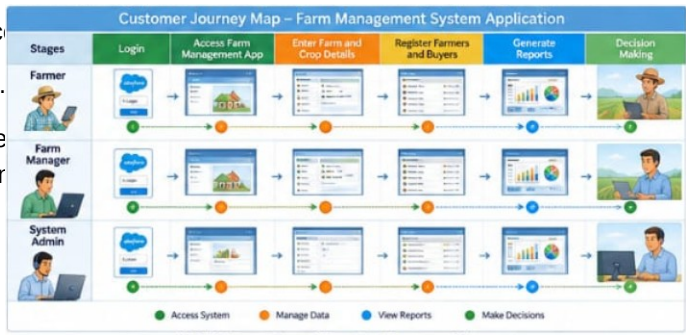
■ Farm details are stored in the Farm object, production information is maintained through the Crop object, farmer details are

recorded through the Farmer object and buyer information is maintained through the Buyer object.

■ The purpose also includes creating relationships between records so information can be reviewed together.

Section relevance: The information in 2. Project Purpose forms part of the Farm Management System documentation and is connected to the project's overall data management, Salesforce configuration, security, reporting, testing or future scope structure as described in this report. This section is intended to make the project easier to review, present and





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3. Project Objectives

The first objective is to maintain centralized and organized records for farms, crops, farmers and buyers. The second objective is to provide structured data entry so information can be stored using appropriate field types. The third objective is to make it easier for users to update and search records without maintaining separate documents.

Another objective is to connect related information through Salesforce relationships so users can move between relevant records. The project also aims to provide secure access using profiles, roles, users and sharing rules. Reports and dashboards are included because stored information should be useful for analysis and monitoring.

The system should support collaboration among users with different responsibilities. Testing is another objective so configured components can be checked. Finally, the project creates a foundation for later mobile, AI, weather, IoT, government scheme and market-price enhancements.

PROJECT DETAILS

■ The first objective is to maintain centralized and organized records for farms, crops, farmers and buyers.

■ The second objective is to provide structured data entry so information can be stored using appropriate field types.

■ The third objective is to make it easier for users to update and search records without maintaining separate documents.

■ Another objective is to connect related information through Salesforce relationships so users can move between relevant records.

Section relevance: The information in 3. Project Objectives forms part of the Farm Management System documentation and is connected to the project's overall data management, Salesforce configuration, security, reporting, testing or future-scope structure as described in this report. This section is intended to make the project easier to review, present and understand.

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4. Problem Statement

Farm management involves information from different areas such as farmers, farms, crops and buyers. If these records are maintained separately, it can become difficult to locate the latest information or connect related details. Farmers need a simple way to maintain their farm information and crop-related details.

Farm managers need access to records that help them monitor farms and production information. Buyers need their contact

and classification details to be maintained accurately. System administrators need a way to organize records and manage access for different users.

Manual records can increase the chance of duplicate entries, missing information and inconsistent updates. The project therefore identifies the need for a centralized platform where major information areas can be maintained together.

Salesforce provides custom objects, relationships, security settings, reports and dashboards that can be configured for this purpose. The problem statement establishes the practical need for the proposed system.

PROJECT DETAILS

■ Farm management involves information from different areas such as farmers, farms, crops and buyers.

■ If these records are maintained separately, it can become difficult to locate the latest information or connect related details.

■ Farmers need a simple way to maintain their farm information and crop-related details.

■ Farm managers need access to records that help them monitor farms and production information.

Section relevance: The information in 4. Problem Statement forms part of the Farm Management System documentation and is connected to the project's overall data management, Salesforce configuration, security, reporting, testing or future-scope structure as described in this report. This section is intended to make the project easier to review, present and understand.

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5. Stakeholders

The Farm Management System considers four primary stakeholder groups. The Farmer is associated with farm information and needs a convenient way to maintain relevant records. The Farm Manager needs access to farm and crop information for monitoring and coordination.

The Buyer is associated with buyer information and requires accurate contact and classification details. The System Administrator is responsible for configuring the Salesforce environment, maintaining access and supporting the overall application. Each stakeholder has different responsibilities, so a single access level may not be suitable for everyone.

The project therefore includes profiles, roles, users and sharing rules as part of its security design. Reports and dashboards

can also support different information needs. Understanding stakeholders helps ensure that the project design remains connected to user responsibilities and provides a basis for planning navigation, data entry, access and reporting requirements.

PROJECT DETAILS

■ The Farm Management System considers four primary stakeholder groups.

■ The Farmer is associated with farm information and needs a convenient way to maintain relevant records.

■ The Farm Manager needs access to farm and crop information for monitoring and coordination.

■ The Buyer is associated with buyer information and requires accurate contact and classification details.

Section relevance: The information in 5. Stakeholders forms part of the Farm Management System documentation and is connected to the project's overall data management, Salesforce configuration, security, reporting, testing or future-scope structure as described in this report. This section is intended to make the project easier to review, present and understand.

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6. Empathy Map

The empathy map describes the information needs and experiences of users working with the farm management process.

Users may see farm records, crop details, farmer information and buyer records during daily activities. They may hear updates from farmers, farm managers or other stakeholders about changes in agricultural information.

Users may think about whether the information they view is current, complete and correctly entered. They may feel that a complicated manual process takes additional time and creates uncertainty. They may say that they need a record quickly or

want a simple way to update information.

Their actions may include entering new records, editing existing records, checking related information and reviewing reports. These observations support the need for a centralized and understandable interface. Salesforce Lightning can provide structured pages and navigation for these activities. The empathy map therefore connects user needs with the features planned in the project.

PROJECT DETAILS

■ The empathy map describes the information needs and experiences of users working with the farm management process.

■ Users may see farm records, crop details, farmer information and buyer records during daily activities.

■ They may hear updates from farmers, farm managers or other stakeholders about changes in agricultural information.

■ Users may think about whether the information they view is current, complete and correctly entered.

Section relevance: The information in 6. Empathy Map forms part of the Farm Management System documentation and is connected to the project's overall data management, Salesforce configuration, security, reporting, testing or future-scope structure as described in this report. This section is intended to make the project easier to review, present and understand.

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7. Brainstorming

The brainstorming stage identifies ideas that can make farm information management more systematic. One idea is to create separate Salesforce custom objects for Farm, Crop, Farmer and Buyer so that each information area has a clear structure. Another idea is to use relationships so users can connect related records.

Controlled picklist values can reduce inconsistent entries for fields such as Farm Type, Soil Quality, Irrigation Type and Buyer Type. Security ideas include creating role-specific profiles and using sharing rules where additional access is required. Reporting ideas include creating a Farmer with Farms report and another report using Crops and Buyers. Dashboard ideas include using a Gauge component for a quick summary. The brainstorming stage also considers user adoption, testing and future integrations. These ideas are later organized into requirements and implementation activities. The result is a practical set of features that can be configured within Salesforce.

PROJECT DETAILS

■ The brainstorming stage identifies ideas that can make farm information management more systematic.

■ One idea is to create separate Salesforce custom objects for Farm, Crop, Farmer and Buyer so that each information area

has a clear structure.

■ Another idea is to use relationships so users can connect related records.

■ Controlled picklist values can reduce inconsistent entries for fields such as Farm Type, Soil Quality, Irrigation Type and Buyer Type.

Section relevance: The information in 7. Brainstorming forms part of the Farm Management System documentation and is connected to the project's overall data management, Salesforce configuration, security, reporting, testing or future-scope structure as described in this report. This section is intended to make the project easier to review, present and understand.

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8. Customer Journey

The customer journey begins when a user identifies agricultural information that needs to be recorded or reviewed. The user enters Salesforce and opens the appropriate application area. The next step is selecting the relevant object or tab, such as Farm, Crop, Farmer or Buyer.

The user enters information using configured fields. Picklists, field types and relationships help structure the data during entry. After the record is saved, the user can review the information and make updates when required.

Related records can be opened to understand connections between farmers, farms and crops. Once sufficient records are available, users can open reports to inspect detailed information. Dashboards can provide summarized views of selected report data. This journey describes how the system supports a user from initial data entry through review and analysis. It also shows why navigation, relationships and analytics are important parts of the overall application experience.

PROJECT DETAILS

■ The customer journey begins when a user identifies agricultural information that needs to be recorded or reviewed.

■ The user enters Salesforce and opens the appropriate application area.

■ The next step is selecting the relevant object or tab, such as Farm, Crop, Farmer or Buyer.

■ The user enters information using configured fields.

Section relevance: The information in 8. Customer Journey forms part of the Farm Management System documentation and is connected to the project's overall data management, Salesforce configuration, security, reporting, testing or future-scope structure as described in this report. This section is intended to make the project easier to review, present and understand.

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9. Requirement Overview

The requirement analysis stage defines what the Farm Management System should provide. Functional requirements focus

on maintaining farmer, farm, crop and buyer information, connecting related records and providing reports and dashboards.

The system also needs security features that support controlled access for different users.

Non-functional requirements focus on qualities such as usability, reliability, maintainability, data consistency and security.

The requirements are connected to the proposed Salesforce configuration so each requirement can be mapped to a platform feature. Custom objects address structured data storage, relationships address connections between records, and reports address detailed information analysis.

Profiles, roles and sharing rules address access management. The requirements also support testing because each major function can be checked after configuration. This stage creates a clear foundation for project design and development activities and keeps implementation connected to the original project goals.

PROJECT DETAILS

■ The requirement analysis stage defines what the Farm Management System should provide.

■ Functional requirements focus on maintaining farmer, farm, crop and buyer information, connecting related records and providing reports and dashboards.

■ The system also needs security features that support controlled access for different users.

■ Non-functional requirements focus on qualities such as usability, reliability, maintainability, data consistency and security.

Section relevance: The information in 9. Requirement Overview forms part of the Farm Management System documentation and is connected to the project's overall data management, Salesforce configuration, security, reporting, testing or future-scope structure as described in this report. This section is intended to make the project easier to review, present and understand.

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10. Functional Requirements

The Farm Management System includes seven major functional requirements. FR-1 is the ability to maintain farmer information using the Farmer object. FR-2 is the ability to maintain farm information using the Farm object. FR-3 is the ability to maintain crop production information using the Crop object.

FR-4 is the ability to maintain buyer information using the Buyer object. FR-5 is the ability to connect related records through Salesforce relationships. FR-6 is the ability to generate reports and dashboards for analysis. FR-7 is the ability to support secure access through Salesforce security configuration.

These requirements describe the core behavior expected from the project. Each requirement is represented by one or more

configured components in the Salesforce environment. The requirements also provide a basis for testing because the project team can verify whether each expected function is available. Together, they define the operational scope of the system.

PROJECT DETAILS

■ The Farm Management System includes seven major functional requirements.

■ FR-1 is the ability to maintain farmer information using the Farmer object.

■ FR-2 is the ability to maintain farm information using the Farm object.

■ FR-3 is the ability to maintain crop production information using the Crop object.

Section relevance: The information in 10. Functional Requirements forms part of the Farm Management System documentation and is connected to the project's overall data management, Salesforce configuration, security, reporting, testing or future-scope structure as described in this report. This section is intended to make the project easier to review, present and understand.

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11. Non-Functional Requirements

The Farm Management System should provide a usable interface so users can navigate between major areas without unnecessary complexity. The system should support security by controlling access according to responsibilities. Reliability is important because users depend on stored records and reports for their work.

Maintainability is required so fields, relationships, reports and access settings can be updated when project needs change. Data consistency is supported through suitable field types and controlled values. The application should organize information clearly so users can understand records and relationships.

The project should provide a stable reporting and dashboard experience within the configured Salesforce environment. These non-functional requirements complement the functional requirements and describe the quality expected from the solution. They are considered during configuration and testing. The result is a project that focuses not only on what the system does, but also on how the system supports users.

PROJECT DETAILS

■ The Farm Management System should provide a usable interface so users can navigate between major areas without unnecessary complexity.

■ The system should support security by controlling access according to responsibilities.

■ Reliability is important because users depend on stored records and reports for their work.

■ Maintainability is required so fields, relationships, reports and access settings can be updated when project needs change.

Section relevance: The information in 11. Non-Functional Requirements forms part of the Farm Management System documentation and is connected to the project's overall data management, Salesforce configuration, security, reporting, testing or future-scope structure as described in this report. This section is intended to make the project easier to review, present and understand.

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12. Data Flow

The data flow begins with a user such as a farmer, farm manager, buyer or administrator entering or reviewing information in Salesforce. The user selects the appropriate object and provides values through the record interface. The information is stored in the corresponding custom object.

Relationships connect records so information can be followed from one object to another. For example, a Farm record can be associated with a Farmer, while a Crop record can be associated with a Farm. Security configuration determines which users can access particular records.

Once records are stored, reports can organize the information into useful views. Dashboard components can then summarize report information for quick monitoring. Users can return to the underlying records when detailed inspection is required. This flow creates a continuous path from data entry to storage, relationship management and analysis and explains how the different Salesforce components work together.

PROJECT DETAILS

■ The data flow begins with a user such as a farmer, farm manager, buyer or administrator entering or reviewing information in

Salesforce.

■ The user selects the appropriate object and provides values through the record interface.

■ The information is stored in the corresponding custom object.

■ Relationships connect records so information can be followed from one object to another.

Section relevance: The information in 12. Data Flow forms part of the Farm Management System documentation and is connected to the project's overall data management, Salesforce configuration, security, reporting, testing or future-scope structure as described in this report. This section is intended to make the project easier to review, present and understand.

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13. Technology Stack

Salesforce is the main technology platform used for the Farm Management System. Salesforce Lightning Experience provides the interface through which users work with the application. Custom objects represent Farm, Crop, Farmer and Buyer information.

Custom fields store the specific values required by each object. Relationships connect records between objects and support

navigation across related information. Profiles provide permission settings for different user groups, while roles support the role hierarchy.

Users are created with suitable profiles and roles, and sharing rules can provide additional record access when required. Reports organize records into detailed analytical views. Dashboards present selected report information visually. Together, these Salesforce features form the technology stack for the project and allow the application to be configured as a centralized agricultural information management system.

PROJECT DETAILS

■ Salesforce is the main technology platform used for the Farm Management System.

■ Salesforce Lightning Experience provides the interface through which users work with the application.

■ Custom objects represent Farm, Crop, Farmer and Buyer information.

■ Custom fields store the specific values required by each object.

Section relevance: The information in 13. Technology Stack forms part of the Farm Management System documentation and is connected to the project's overall data management, Salesforce configuration, security, reporting, testing or future-scope structure as described in this report. This section is intended to make the project easier to review, present and understand.

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14. Problem-Solution Fit

The project maps identified problems to specific Salesforce capabilities. Scattered agricultural records are addressed through centralized custom objects. Difficult data entry is addressed through structured fields, picklists and suitable field types.

Relationships address the need to connect farmer, farm and crop information. Access concerns are addressed through profiles, roles, users and sharing rules. Limited visibility into stored information is addressed through reports and dashboards.

Manual work can be reduced because users can maintain records within one Salesforce environment instead of relying on separate documents. The proposed solution also supports future expansion because additional Salesforce configuration or integrations can be added later. This problem-solution fit shows how project requirements translate into platform features. It also provides a logical connection between the problem statement, proposed solution and implementation plan.

PROJECT DETAILS

■ The project maps identified problems to specific Salesforce capabilities.

■ Scattered agricultural records are addressed through centralized custom objects.

■ Difficult data entry is addressed through structured fields, picklists and suitable field types.

■ Relationships address the need to connect farmer, farm and crop information.

Section relevance: The information in 14. Problem-Solution Fit forms part of the Farm Management System documentation and is connected to the project's overall data management, Salesforce configuration, security, reporting, testing or future-scope structure as described in this report. This section is intended to make the project easier to review, present and understand.

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15. Proposed Solution

The proposed solution is a Salesforce Lightning application built around four custom objects: Farm, Crop, Farmer and Buyer. The Farmer object maintains farmer information, while the Farm object maintains farm details and connects the farm

to its farmer. The Crop object stores production-related information and uses a Lookup relationship to the Farm object. The Buyer object maintains buyer contact and classification information. Custom tabs provide direct navigation to these objects. A Lightning App brings the main navigation items together for users.

Fields and relationships provide the underlying data structure. Profiles, roles, users and sharing rules provide the access-control layer. Reports and dashboards provide the analytical layer. This solution is designed to keep the main agricultural information areas together while separating them into logical objects. The result is a structured Salesforce application that supports both record management and analysis.

PROJECT DETAILS

■ The proposed solution is a Salesforce Lightning application built around four custom objects: Farm, Crop, Farmer and Buyer.

■ The Farmer object maintains farmer information, while the Farm object maintains farm details and connects the farm to its

farmer.

■ The Crop object stores production-related information and uses a Lookup relationship to the Farm object.

■ The Buyer object maintains buyer contact and classification information.

Section relevance: The information in 15. Proposed Solution forms part of the Farm Management System documentation and is connected to the project's overall data management, Salesforce configuration, security, reporting, testing or future-scope structure as described in this report. This section is intended to make the project easier to review, present and understand.

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16. Solution Architecture

The solution architecture can be viewed as several connected layers. The first layer contains users such as farmers, farm managers, buyers and administrators. The second layer is the Salesforce Lightning App that provides the main user interface and navigation.

The third layer contains the four custom objects: Farm, Crop, Farmer and Buyer. The fourth layer contains fields and relationships that define how information is stored and connected. The security layer contains profiles, roles, users and sharing rules that determine access.

The analytics layer contains reports and dashboards that use stored records for analysis. These layers work together rather

than operating as separate components. A user enters or updates information through the interface, Salesforce stores the record, relationships connect relevant data, and security controls access. Reports and dashboards then make the stored information useful for monitoring and review.

PROJECT DETAILS

■ The solution architecture can be viewed as several connected layers.

■ The first layer contains users such as farmers, farm managers, buyers and administrators.

■ The second layer is the Salesforce Lightning App that provides the main user interface and navigation.

■ The third layer contains the four custom objects: Farm, Crop, Farmer and Buyer.

Section relevance: The information in 16. Solution Architecture forms part of the Farm Management System documentation and is connected to the project's overall data management, Salesforce configuration, security, reporting, testing or future-scope structure as described in this report. This section is intended to make the project easier to review, present and understand.

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17. Project Planning

Project planning divides implementation into manageable stages. The first stage is requirement analysis, where the project scope, users and information needs are identified. The second stage is object setup, where Farm, Crop, Farmer and Buyer objects are created.

The third stage is interface setup, including custom tabs and the Lightning App. The fourth stage is data-model configuration, where fields and relationships are created. The fifth stage is security configuration, including profiles, roles, users and sharing rules.

The sixth stage is analytics, where reports and dashboards are created. Testing is performed after the major components are configured. This staged approach helps reduce confusion because each part of the system is developed in a logical sequence. It also creates checkpoints where the project can be reviewed before moving to the next activity. The plan supports both implementation and academic project tracking.

PROJECT DETAILS

■ Project planning divides implementation into manageable stages.

■ The first stage is requirement analysis, where the project scope, users and information needs are identified.

■ The second stage is object setup, where Farm, Crop, Farmer and Buyer objects are created.

■ The third stage is interface setup, including custom tabs and the Lightning App.

Section relevance: The information in 17. Project Planning forms part of the Farm Management System documentation and is connected to the project's overall data management, Salesforce configuration, security, reporting, testing or future-scope structure as described in this report. This section is intended to make the project easier to review, present and understand.

Review note: The project terminology and scope are retained consistently across the documentation. The section can be used as part of the final project explanation and demonstration.

Farm Management System: This page forms part of the same project documentation and supports the overall Salesforce-based management, configuration, security, reporting or testing flow.

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18. Product Backlog

The product backlog contains the main tasks required to complete the Farm Management System. Initial tasks include creating the Farm, Crop, Farmer and Buyer custom objects. The next tasks include creating custom tabs and building the Lightning App navigation.

Field configuration follows object creation so each record contains the required information. Relationships are then configured to connect the data model. Security tasks include creating profiles, configuring the role hierarchy, creating users and setting sharing rules.

Analytics tasks include creating reports and the dashboard. Testing tasks verify that the configured features work correctly. The backlog provides a simple checklist that helps track project progress. It also makes it easier to identify which configuration activities are completed and which remain. The final backlog items focus on testing and project review before submission or demonstration.

PROJECT DETAILS

■ The product backlog contains the main tasks required to complete the Farm Management System.

■ Initial tasks include creating the Farm, Crop, Farmer and Buyer custom objects.

■ The next tasks include creating custom tabs and building the Lightning App navigation.

■ Field configuration follows object creation so each record contains the required information.

Section relevance: The information in 18. Product Backlog forms part of the Farm Management System documentation and is connected to the project's overall data management, Salesforce configuration, security, reporting, testing or future-scope structure as described in this report. This section is intended to make the project easier to review, present and understand.

Review note: The project terminology and scope are retained consistently across the documentation. The section can be used as part of the final project explanation and demonstration.

Farm Management System: This page forms part of the same project documentation and supports the overall Salesforce-based management, configuration, security, reporting or testing flow.

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19. Sprint Plan

The project can be divided into six practical sprints. Sprint 1 focuses on requirements, problem understanding and project design. Sprint 2 focuses on creating the four main custom objects: Farm, Crop, Farmer and Buyer.

Sprint 3 focuses on custom tabs and the Lightning App navigation. Sprint 4 focuses on fields and relationships so the data model is complete. Sprint 5 focuses on security, including profiles, role hierarchy, users and sharing rules.

Sprint 6 focuses on reports, dashboards and testing. Each sprint has a defined group of activities so the project can progress step by step. The sprint structure also provides a way to review the work completed before continuing. This approach helps keep the Salesforce configuration organized and makes project documentation easier to prepare and explain during review.

PROJECT DETAILS

■ The project can be divided into six practical sprints.

■ Sprint 1 focuses on requirements, problem understanding and project design.

■ Sprint 2 focuses on creating the four main custom objects: Farm, Crop, Farmer and Buyer.

■ Sprint 3 focuses on custom tabs and the Lightning App navigation.

Section relevance: The information in 19. Sprint Plan forms part of the Farm Management System documentation and is connected to the project's overall data management, Salesforce configuration, security, reporting, testing or future-scope structure as described in this report. This section is intended to make the project easier to review, present and understand.

Review note: The project terminology and scope are retained consistently across the documentation. The section can be used as part of the final project explanation and demonstration.

Farm Management System: This page forms part of the same project documentation and supports the overall Salesforce-based management, configuration, security, reporting or testing flow.

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20. Salesforce Platform

Salesforce provides the cloud platform used to configure the Farm Management System. The platform includes Setup tools for creating objects, fields, relationships, profiles, roles and users. Lightning Experience provides the user interface for working with records and applications.

Object Manager is used to configure custom objects and their fields. Custom tabs provide navigation to object records. The Lightning App brings important navigation items into one application.

Reports allow users to organize records into detailed views. Dashboards use report information to provide visual summaries. Security settings control how users interact with records. Because these capabilities are available within Salesforce, the project can be implemented without creating a separate standalone database application for the core requirements. The platform therefore provides both the development environment and the user-facing application foundation.

PROJECT DETAILS

■ Salesforce provides the cloud platform used to configure the Farm Management System.

■ The platform includes Setup tools for creating objects, fields, relationships, profiles, roles and users.

■ Lightning Experience provides the user interface for working with records and applications.

■ Object Manager is used to configure custom objects and their fields.

Section relevance: The information in 20. Salesforce Platform forms part of the Farm Management System documentation and is connected to the project's overall data management, Salesforce configuration, security, reporting, testing or future-scope structure as described in this report. This section is intended to make the project easier to review, present and understand.

Review note: The project terminology and scope are retained consistently across the documentation. The section can be used as part of the final project explanation and demonstration.

Farm Management System: This page forms part of the same project documentation and supports the overall Salesforce-based management, configuration, security, reporting or testing flow.

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21. Developer Org Activity

The Salesforce Developer Org acts as the workspace for building and testing the Farm Management System. The project configuration is performed inside this environment. Custom objects are created first so the main data structure is available. Fields are then added to capture required information for each object. Relationships connect records and support navigation across the data model. Custom tabs and the Lightning App provide the user interface structure.

Profiles, roles and users are configured to support different responsibilities. Sharing rules can be added where additional access is required. Reports and dashboards are created after enough structure is available for meaningful analysis. Testing

is performed within the same environment to verify that the configuration behaves as expected. This sequence keeps development organized and provides a clear path from object creation to final analytics.

PROJECT DETAILS

■ The Salesforce Developer Org acts as the workspace for building and testing the Farm Management System.

■ The project configuration is performed inside this environment.

■ Custom objects are created first so the main data structure is available.

■ Fields are then added to capture required information for each object.

Section relevance: The information in 21. Developer Org Activity forms part of the Farm Management System documentation and is connected to the project's overall data management, Salesforce configuration, security, reporting, testing or future-scope structure as described in this report. This section is intended to make the project easier to review, present and understand.

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22. Farm Object

The Farm custom object stores information about individual farms. The object contains information that describes the type of farming activity, soil quality, irrigation method, location and size. Farm Type is configured as a picklist with Crop, Livestock and Mixed values.

Soil Quality includes Organic, Cover Crops and Composting as defined in the project design. Irrigation Type includes Sprinkler irrigation, Drip irrigation and localized irrigation. Location and Size are maintained as fields for basic farm information.

The Farm object is connected to the Farmer object through a Master-Detail relationship. This relationship helps identify the farmer associated with a farm record. The object provides the main location for farm-level information in the system. Its fields and relationship form an important part of the project data model and support later reporting and analysis.

PROJECT DETAILS

■ The Farm custom object stores information about individual farms.

■ The object contains information that describes the type of farming activity, soil quality, irrigation method, location and size.

■ Farm Type is configured as a picklist with Crop, Livestock and Mixed values.

■ Soil Quality includes Organic, Cover Crops and Composting as defined in the project design.

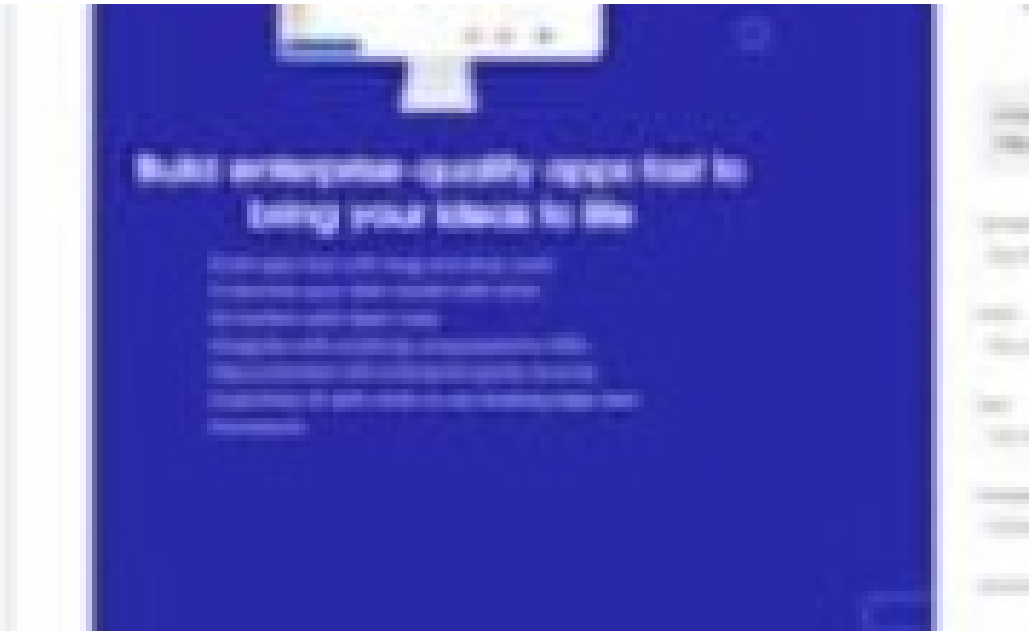
Section relevance: The information in 22. Farm Object forms part of the Farm Management System documentation and is connected to the project's overall data management, Salesforce configuration, security, reporting, testing or future-scope structure as described in this report. This section is intended to make the project easier to review, present and understand.

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23. Farm Object – Field Design

The Farm object field design focuses on capturing information needed to describe a farm consistently. Farm Type uses a picklist so users can choose Crop, Livestock or Mixed rather than entering different spellings. Soil Quality also uses controlled values such as Organic, Cover Crops and Composting.

Irrigation Type provides the project-defined options Sprinkler irrigation, Drip irrigation and localized irrigation. Location provides a place for farm location information, while Size records farm size. The relationship to Farmer provides a connection between the farm and the person responsible for it.

These fields create a clear structure for farm records. They also make information easier to use in reports because values follow defined formats. The field design therefore supports consistency, relationships and later analysis while keeping the Farm object aligned with the requirements of the project.

PROJECT DETAILS

■ The Farm object field design focuses on capturing information needed to describe a farm consistently.

■ Farm Type uses a picklist so users can choose Crop, Livestock or Mixed rather than entering different spellings.

■ Soil Quality also uses controlled values such as Organic, Cover Crops and Composting.

■ Irrigation Type provides the project-defined options Sprinkler irrigation, Drip irrigation and localized irrigation.

Section relevance: The information in 23. Farm Object – Field Design forms part of the Farm Management System documentation and is connected to the project's overall data management, Salesforce configuration, security, reporting, testing or future-scope structure as described in this report. This section is intended to make the project easier to review, present and understand.

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24. Crop Object

The Crop custom object stores information about crop production activities. Crop Name identifies the crop record, while Planting Date and Harvest Date capture important production dates. The Farm field is configured as a Lookup relationship so a crop can be associated with a farm.

Actual Yield stores the production value using the configured currency field. Total Acres Harvested stores the number of acres associated with the harvest. The object also includes a formula field for Expected Yield per Acre.

These fields provide a basic structure for recording crop production information. By connecting Crop to Farm, users can understand which farm contains the crop record. The Crop object therefore extends farm information into production-related

details. It also provides data that can later be included in reports and analytical views.

PROJECT DETAILS

■ The Crop custom object stores information about crop production activities.

■ Crop Name identifies the crop record, while Planting Date and Harvest Date capture important production dates.

■ The Farm field is configured as a Lookup relationship so a crop can be associated with a farm.

■ Actual Yield stores the production value using the configured currency field.

Section relevance: The information in 24. Crop Object forms part of the Farm Management System documentation and is connected to the project's overall data management, Salesforce configuration, security, reporting, testing or future-scope structure as described in this report. This section is intended to make the project easier to review, present and understand.

Review note: The project terminology and scope are retained consistently across the documentation. The section can be u

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25. Crop – Formula and Measurement

The Crop object includes a formula field named Expected Yield per Acre. The project design uses Actual Yield divided by Total Acres Harvested to derive this value. A formula field is useful because the calculated result can be generated from stored values instead of entered manually each time.

The Actual Yield field provides the production value, while Total Acres Harvested provides the measurement used in the calculation. The formula connects these two fields within the crop record. This approach can reduce repeated manual calculation and provide a consistent derived value.

Users can review the calculated result along with the original production information. The formula can also be used in reports and analysis. Proper data entry for the underlying fields remains important because the formula depends on those values. This feature demonstrates how Salesforce can derive useful information from stored project data.

PROJECT DETAILS

■ The Crop object includes a formula field named Expected Yield per Acre.

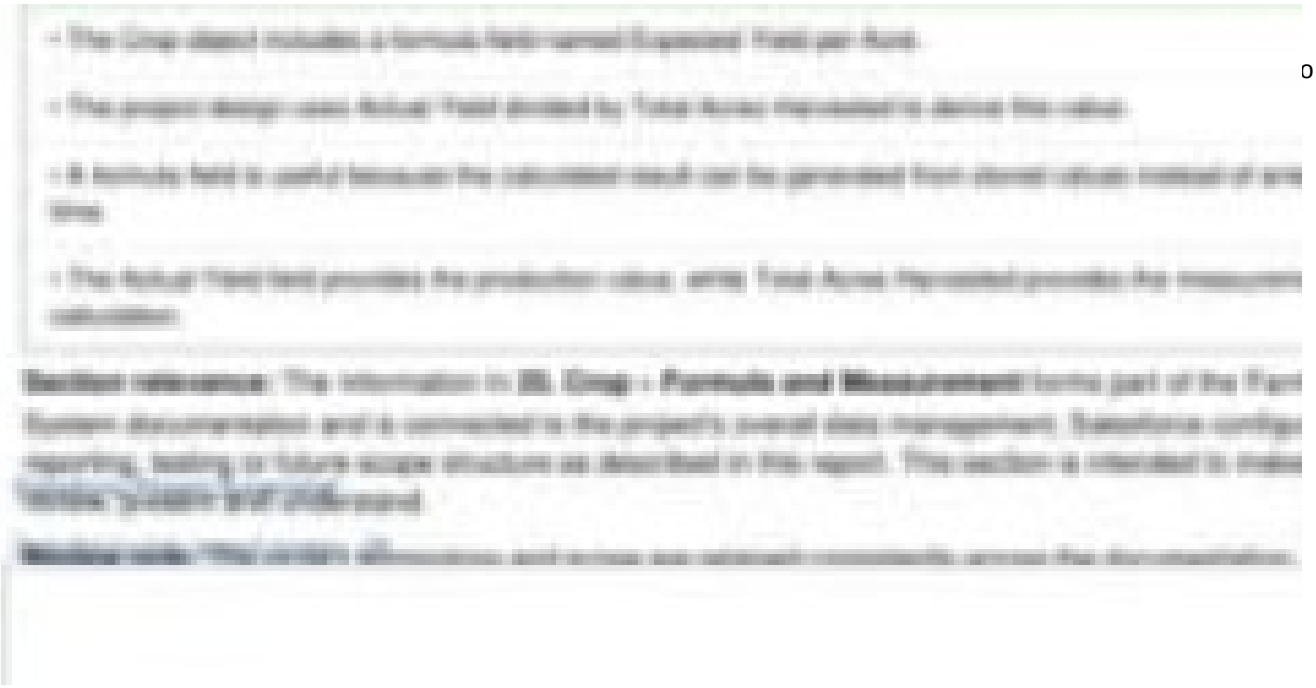
■ The project design uses Actual Yield divided by Total Acres Harvested to derive this value.

■ A formula field is useful because the calculated result can be generated from stored values instead of entered manually each

time.

■ The Actual Yield field provides the production value, while Total Acres Harvested provides the measurement used in the calculation.

Section relevance: The information in 25. Crop – Formula and Measurement forms part of the Farm Management



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26. Farmer Object

The Farmer custom object stores information about farmers who are associated with farms in the project. Farmer Name identifies the record, while Contact Number provides a phone field for communication details. State, District and Village are maintained as picklists to organize location information.

The project also uses field dependency so location selections can be controlled. This structure reduces the need for free-form location entry and helps maintain consistent values. The Farmer object is connected to the Farm object through the project relationship design.

This allows farm records to be associated with a farmer. The object provides the person-related part of the main agricultural

data model. Its fields can also be used in reports to show farmers together with their related farm information. The Farmer object therefore supports both record management and connected reporting.

PROJECT DETAILS

■ The Farmer custom object stores information about farmers who are associated with farms in the project.

■ Farmer Name identifies the record, while Contact Number provides a phone field for communication details.

■ State, District and Village are maintained as picklists to organize location information.

■ The project also uses field dependency so location selections can be controlled.

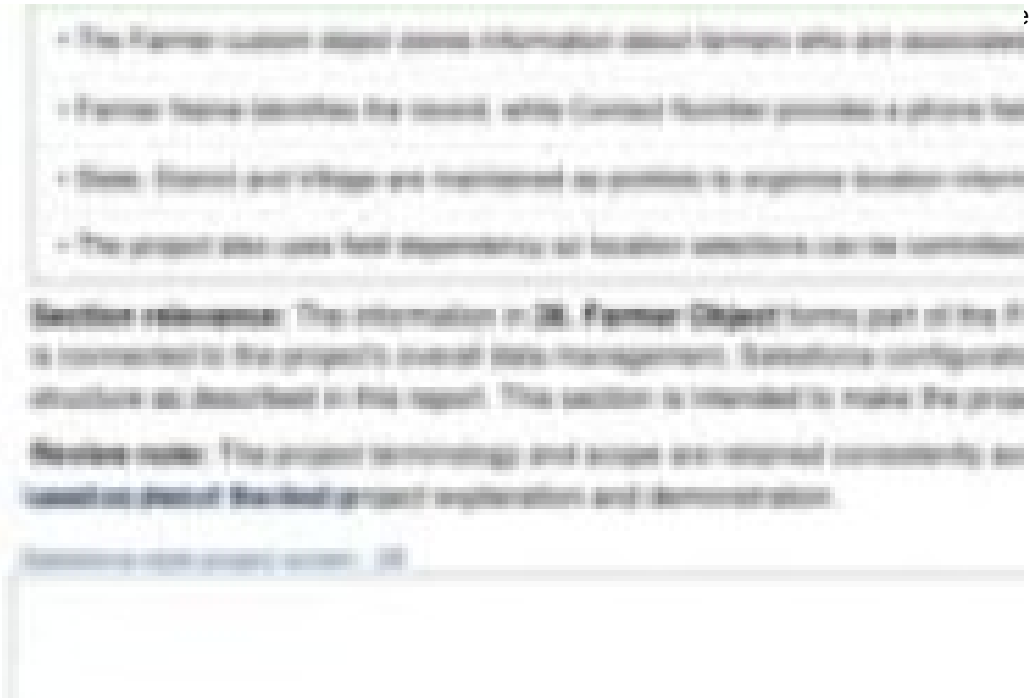
Section relevance: The information in 26. Farmer Object forms part of the Farm Management System documentation and is connected to the project's overall data management, Salesforce configuration, security, reporting, testing or future-scope structure as described in this report. This section is intended to make the project easier to review, present and understand.

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27. Farmer – Field Dependency

The Farmer object uses State, District and Village fields to organize location information. These fields are configured as picklists rather than unrestricted text fields. The project design includes a field dependency so available dependent values can be controlled based on the selected controlling value.

An example included in the project design is Telangana followed by Yadadri Bhongir at the district level. Field dependency helps guide users toward valid combinations of location information. It can also improve consistency when records are entered by different users.

The purpose is not to create a complete geographic database, but to provide controlled location selection within the project.

This configuration is part of the data-quality design. It also provides an example of how Salesforce field configuration can support structured information entry and make location data easier to maintain.

PROJECT DETAILS

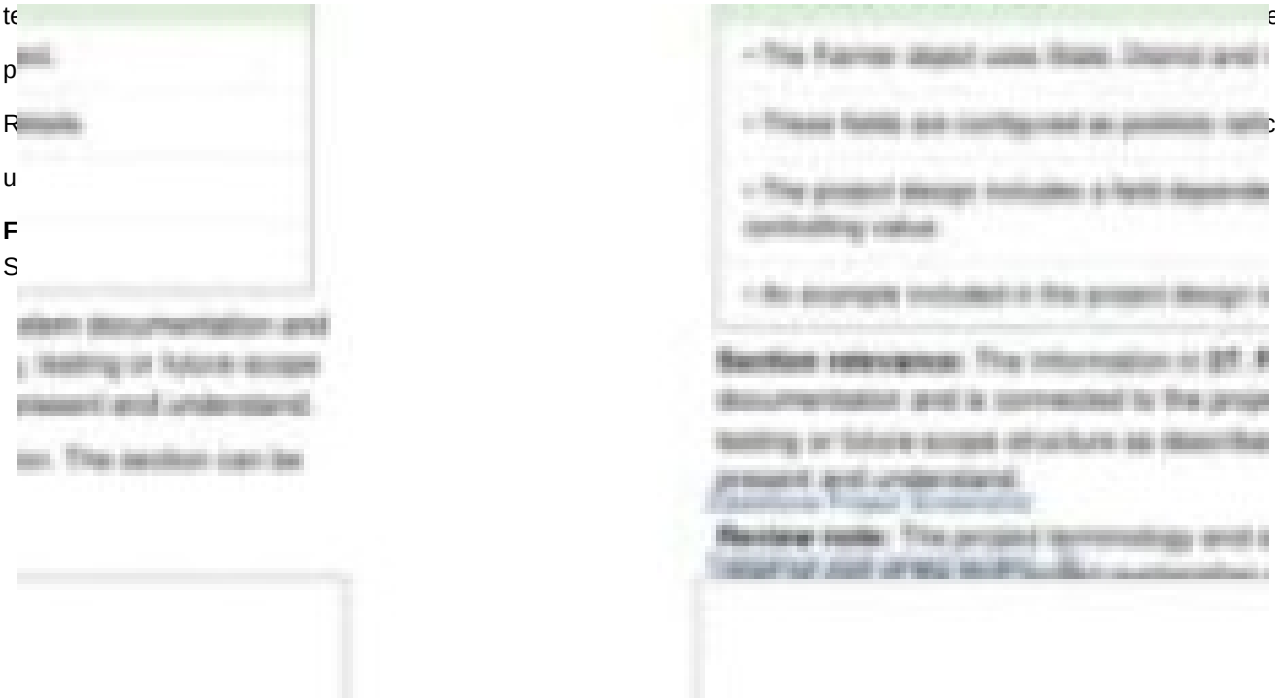
■ The Farmer object uses State, District and Village fields to organize location information.

■ These fields are configured as picklists rather than unrestricted text fields.

■ The project design includes a field dependency so available dependent values can be controlled based on the selected controlling value.

■ An example included in the project design is Telangana followed by Yadadri Bhongir at the district level.

Section relevance: The information in 27. Farmer – Field Dependency forms part of the Farm Management System documentation and is connected to the project's overall data management, Salesforce configuration, security, reporting, te eview, p R can be u **F** S



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28. Buyer Object

The Buyer custom object stores information about buyers relevant to the farm management process. Buyer Name identifies

the buyer record. Contact is configured as a Phone field so a telephone number can be stored in an appropriate format. Email is configured as an Email field for buyer communication information. Address is maintained using a Text Area field so address information can be recorded. Buyer Type is configured as a picklist with Wholesale and Retail values.

These fields provide a simple but structured buyer record. The Buyer object keeps buyer information separate from farmer and farm records while still allowing it to be included in reports. The object is also relevant to the sharing design because Buyer records are shared with the Sales Department in the project configuration. This gives the buyer area a clear role in the application data model.

PROJECT DETAILS

■ The Buyer custom object stores information about buyers relevant to the farm management process.

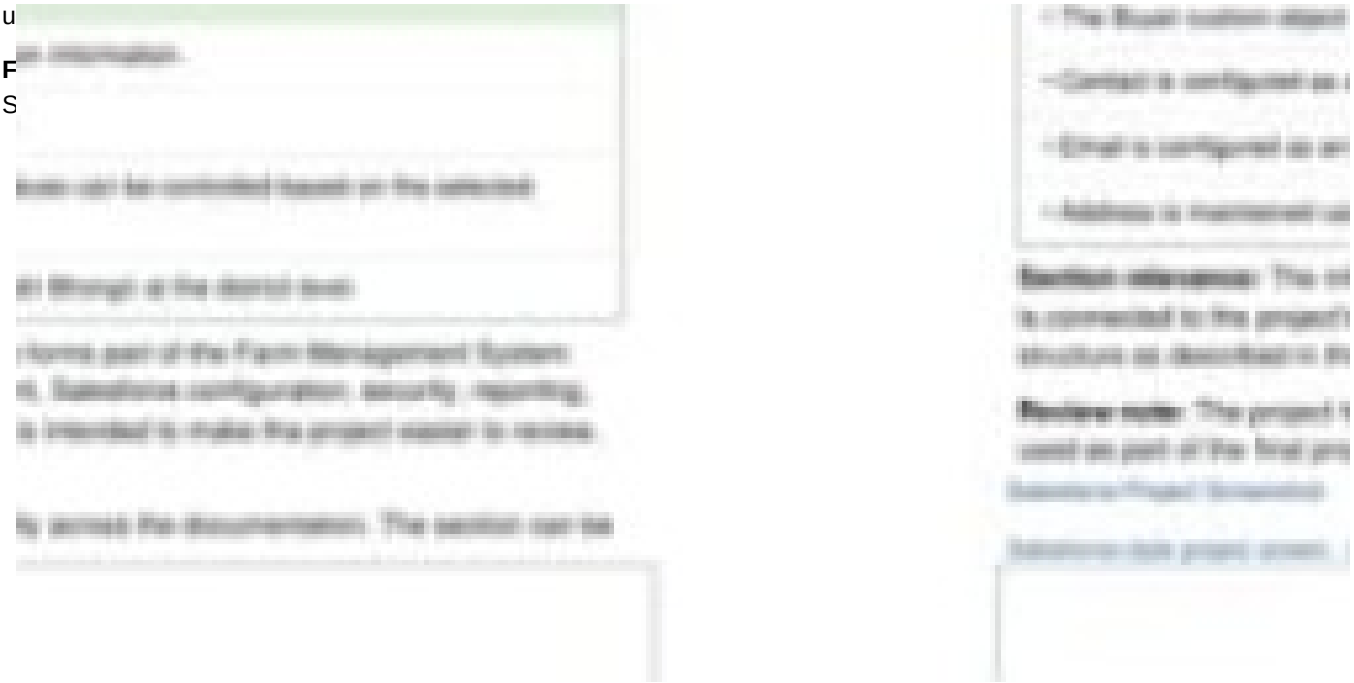
■ Contact is configured as a Phone field so a telephone number can be stored in an appropriate format.

■ Email is configured as an Email field for buyer communication information.

■ Address is maintained using a Text Area field so address information can be recorded.

Section relevance: The information in 28. Buyer Object forms part of the Farm Management System documentation and is connected to the project's overall data management, Salesforce configuration, security, reporting, testing or future-scope structure as described in this report. This section is intended to make the project easier to review, present and understand.

Review note: The project terminology and scope are retained consistently across the documentation. The section can be u



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29. Buyer – Classification

Buyer Type is an important field in the Buyer object because it provides a simple classification of buyers. The project defines Wholesale and Retail as the available values. A picklist is suitable for this classification because users select from a

controlled set of options rather than entering different descriptions.

Consistent classification can make buyer records easier to filter and group in reports. The remaining buyer fields provide contact and address information that supports the record. The Buyer object can therefore be used as a structured source for buyer-related reporting.

Its access is also considered in the sharing rules, where Buyer records are shared with the Sales Department using Read/Write access. This configuration supports collaboration while keeping buyer information organized. The classification field is a small but useful part of the overall data model and reporting structure.

PROJECT DETAILS

■ Buyer Type is an important field in the Buyer object because it provides a simple classification of buyers.

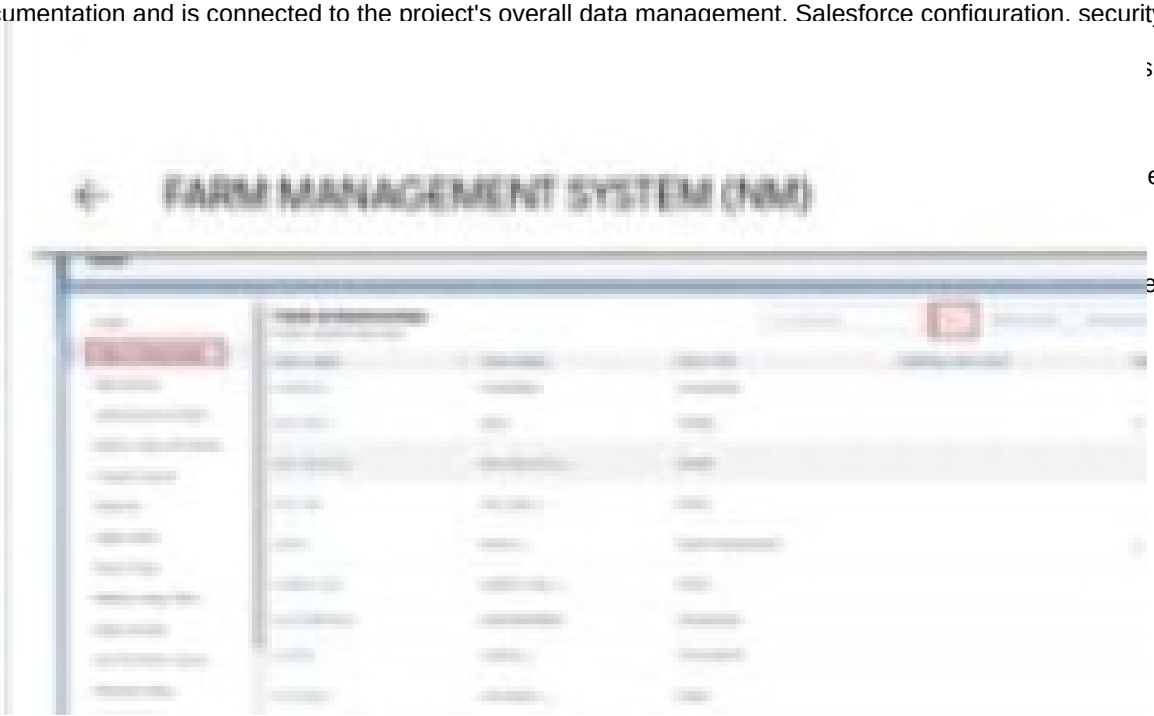
■ The project defines Wholesale and Retail as the available values.

■ A picklist is suitable for this classification because users select from a controlled set of options rather than entering different

descriptions.

■ Consistent classification can make buyer records easier to filter and group in reports.

Section relevance: The information in 29. Buyer – Classification forms part of the Farm Management System documentation and is connected to the project's overall data management, Salesforce configuration, security, reporting, te sier to review, p R ection can be u **F** erall S



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30. Custom Tabs

Custom tabs provide a simple navigation method for the Farm Management System. The project includes tabs for Farm, Crop, Farmer and Buyer objects. Each tab gives users a direct way to open the corresponding records and related list views.

Tabs are useful because they make the main information areas visible from the application navigation. They also help users

avoid repeatedly searching through setup pages to reach an object. The custom tabs are included in the Lightning App so the project has a single user-facing navigation structure.

This configuration connects the data model to the application interface. When users open a tab, they can create, view or update records according to their permissions. Custom tabs therefore support usability, navigation and user adoption within the project.

PROJECT DETAILS

■ Custom tabs provide a simple navigation method for the Farm Management System.

■ The project includes tabs for Farm, Crop, Farmer and Buyer objects.

■ Each tab gives users a direct way to open the corresponding records and related list views.

■ Tabs are useful because they make the main information areas visible from the application navigation.

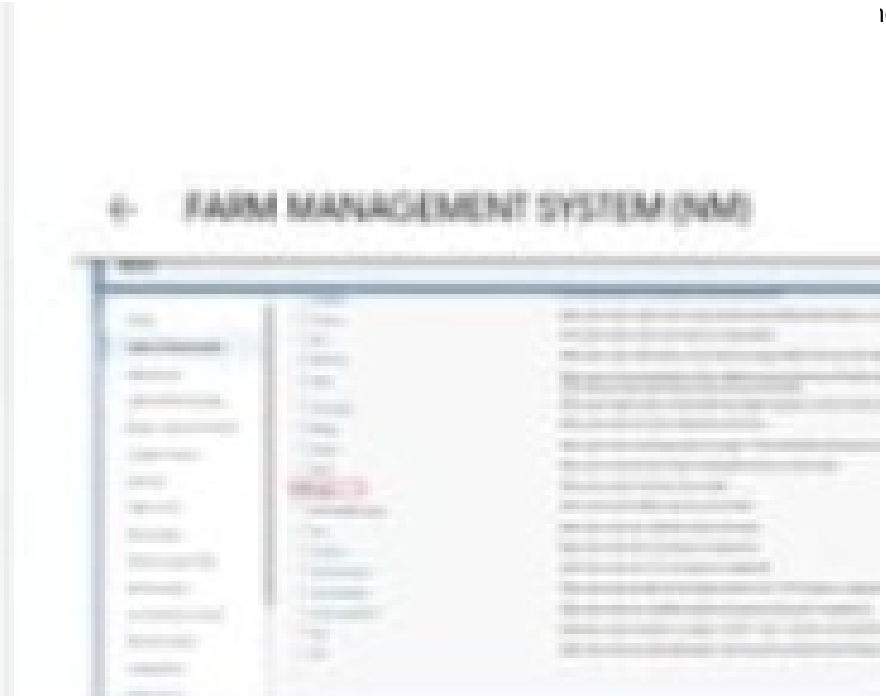
Section relevance: The information in 30. Custom Tabs forms part of the Farm Management System documentation and is connected to the project's overall data management, Salesforce configuration, security, reporting, testing or future-scope structure as described in this report. This section is intended to make the project easier to review, present and understand.

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31. Lightning App

The Lightning App brings the main components of the Farm Management System into one navigation structure. The application can include Home, Farms, Crops, Farmers, Buyers, Reports and Dashboards. This arrangement places major record areas and analytics features together.

Users can move between object tabs and analytical views without leaving the application context. The Lightning App is important because a good data model still needs an accessible interface. The navigation structure should reflect the way users work with the project.

Farm and Crop records support agricultural data management, while Farmer and Buyer records support people and buyer information. Reports and Dashboards provide the analytical part of the application. Together, these navigation items create a simple user-facing structure and provide a practical starting point for daily use.

PROJECT DETAILS

■ The Lightning App brings the main components of the Farm Management System into one navigation structure.

■ The application can include Home, Farms, Crops, Farmers, Buyers, Reports and Dashboards.

■ This arrangement places major record areas and analytics features together.

■ Users can move between object tabs and analytical views without leaving the application context.

Section relevance: The information in 31. Lightning App forms part of the Farm Management System documentation and is connected to the project's overall data management, Salesforce configuration, security, reporting, testing or future-scope structure as described in this report. This section is intended to make the project easier to review, present and understand.

Review note: The project terminology and scope are retained consistently across the documentation. The section can be u

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32. Fields and Relationships

Fields define the individual pieces of information stored in each Salesforce object, while relationships define how records in different objects are connected. The Farm object contains farm-specific fields and is connected to Farmer through a Master-Detail relationship in the project design. The Crop object contains production fields and uses a Lookup relationship to Farm.

The Farmer object contains contact and location fields, while the Buyer object contains contact and classification information. Relationships allow users to move between relevant records and support combined reporting. Field types such as Phone, Email, Date, Currency, Number and Picklist provide structure to stored values.

This combination of fields and relationships forms the data model of the application. It also supports the project's goal of keeping agricultural information organized, connected and suitable for analysis.

PROJECT DETAILS

■ Fields define the individual pieces of information stored in each Salesforce object, while relationships define how records in

different objects are connected.

■ The Farm object contains farm-specific fields and is connected to Farmer through a Master-Detail relationship in the project

design.

■ The Crop object contains production fields and uses a Lookup relationship to Farm.

■ The Farmer object contains contact and location fields, while the Buyer object contains contact and classification information.



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33. Relationship Design

The relationship design connects the main objects in the Farm Management System. Farm has a Master-Detail relationship

with Farmer according to the project design, which links farm records to their related farmer. Crop uses a Lookup

relationship to Farm so that a crop record can reference the farm where the crop information belongs.

These relationships support navigation and reporting because related information can be viewed together. A relationship

also reduces the need to repeat the same identifying information in multiple records. For example, the Farm relationship

allows a crop record to refer to its associated farm instead of storing a separate copy of all farm details.

The relationship design is an important part of the system architecture. It also demonstrates how Salesforce objects can

work together as a connected data model for the project.

PROJECT DETAILS

■ The relationship design connects the main objects in the Farm Management System.

■ Farm has a Master-Detail relationship with Farmer according to the project design, which links farm records to their related

farmer.

■ Crop uses a Lookup relationship to Farm so that a crop record can reference the farm where the crop information belongs.

■ These relationships support navigation and reporting because related information can be viewed together.

Section relevance: The information in 33. Relationship Design forms part of the Farm Management System

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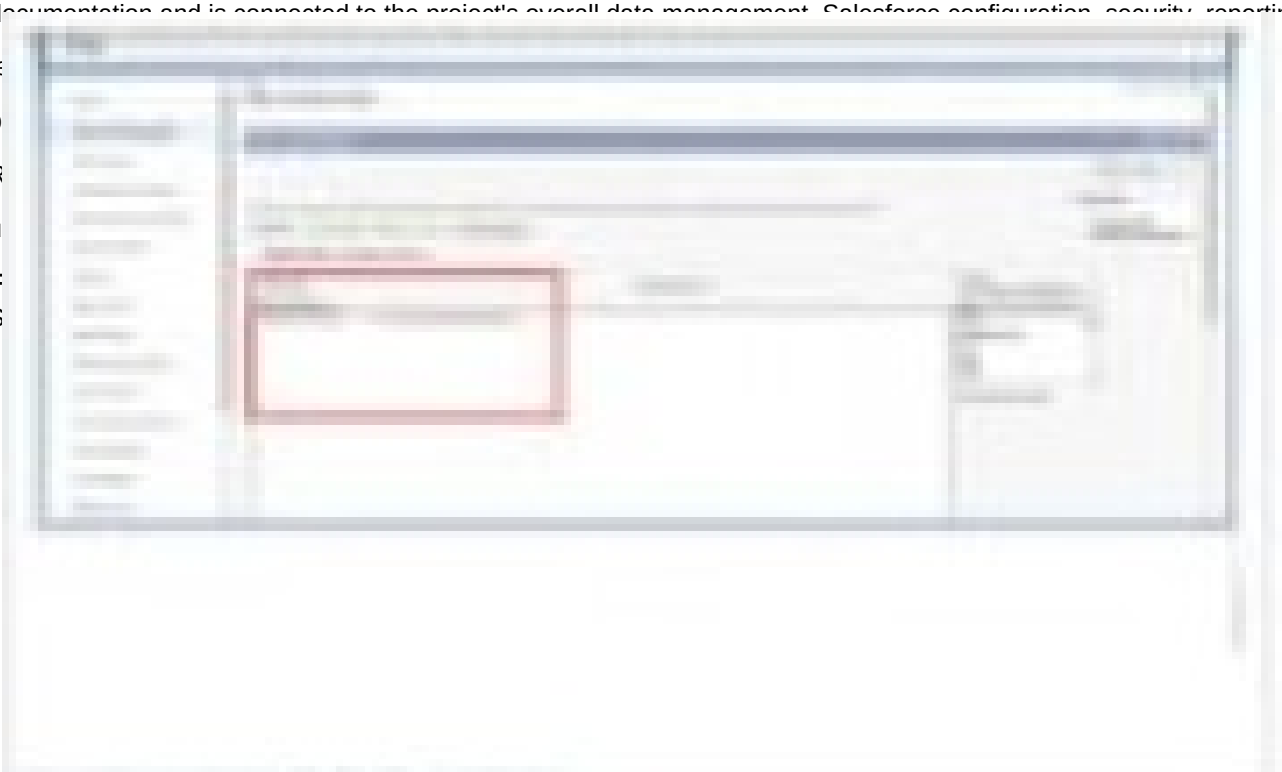
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34. Profiles

Profiles are used to control what Salesforce users can access and what actions they can perform. The project design includes cloning the Standard User profile to create role-specific profiles. One profile is created for Farm Manager responsibilities, while another is created for Sales Department responsibilities.

Cloning provides a starting point from which required permissions can be configured. The profile configuration should reflect

the responsibilities of each user group. Profiles work together with roles and sharing rules, so they should be considered as one part of the overall security model.

The project uses profiles to support controlled access rather than giving every user the same configuration. This approach helps separate responsibilities within the application. Profiles are therefore an important security component of the Farm Management System and should be checked during user-access testing.

PROJECT DETAILS

■ Profiles are used to control what Salesforce users can access and what actions they can perform.

■ The project design includes cloning the Standard User profile to create role-specific profiles.

■ One profile is created for Farm Manager responsibilities, while another is created for Sales Department responsibilities.

■ Cloning provides a starting point from which required permissions can be configured.

Section relevance: The information in 34. Profiles forms part of the Farm Management System documentation and is connected to the project's overall data management, Salesforce configuration, security, reporting, testing or future-scope structure as described in this report. This section is intended to make the project easier to review, present and understand.

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35. Role Hierarchy

The role hierarchy represents the organizational relationship used by the project. The project design places the Farm Manager role under the CEO. A role hierarchy can help organize users and support record visibility according to organizational responsibilities.

In this project, the Farm Manager role provides a clear position for the user responsible for farm-related management. The hierarchy is separate from the profile because the profile controls permissions while the role is part of the organizational access structure. Together with sharing rules and other settings, the role contributes to the overall access model.

Creating the hierarchy is therefore part of the security configuration stage. It also helps the project demonstrate how Salesforce can represent organizational responsibilities and provide a structured foundation for user access.

PROJECT DETAILS

■ The role hierarchy represents the organizational relationship used by the project.

■ The project design places the Farm Manager role under the CEO.

■ A role hierarchy can help organize users and support record visibility according to organizational responsibilities.

■ In this project, the Farm Manager role provides a clear position for the user responsible for farm-related management.

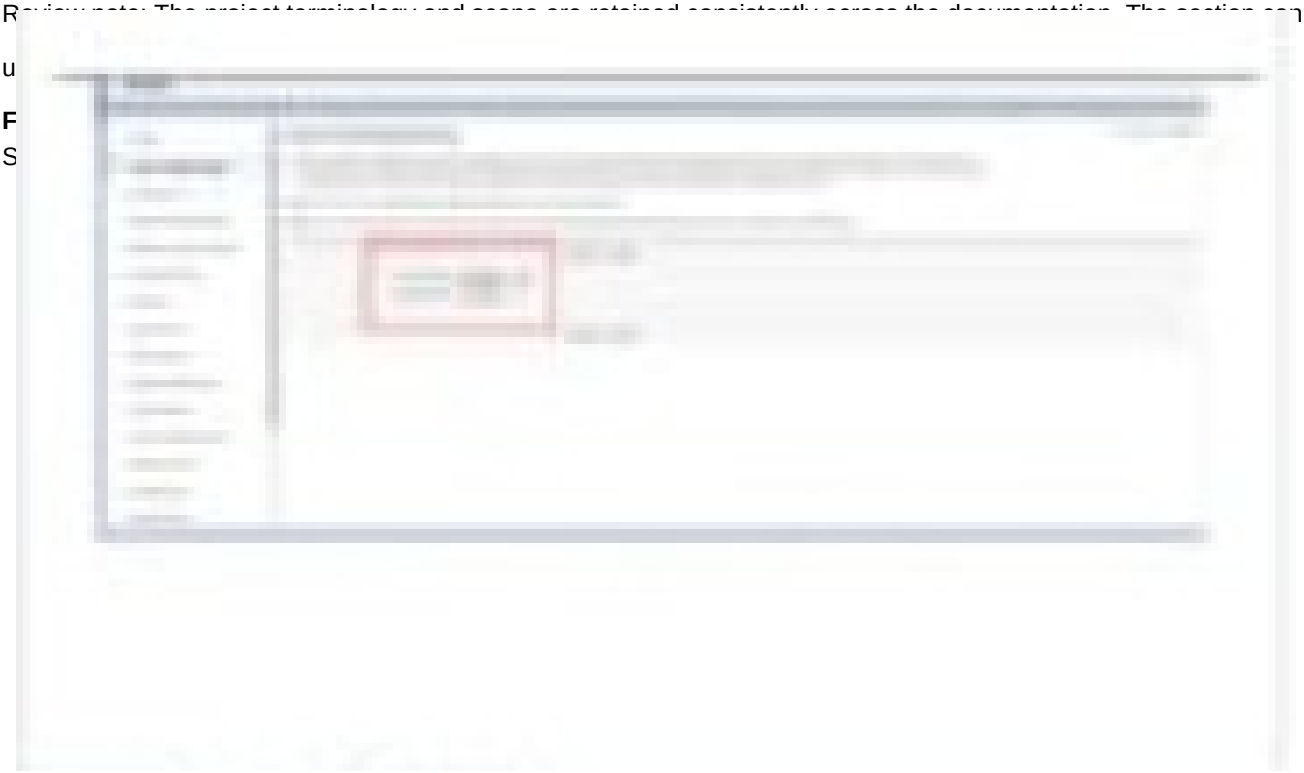
Section relevance: The information in 35. Role Hierarchy forms part of the Farm Management System documentation and is connected to the project's overall data management, Salesforce configuration, security, reporting, testing or future-scope structure as described in this report. This section is intended to make the project easier to review, present and understand.

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36. Users

User records connect individual people to the configured Salesforce access model. The project includes a Farm Manager user with the Standard Platform User license, Farm Manager profile and Farm Manager role. The project also includes a Sales Department user associated with a sales-related profile and responsibility.

Assigning a profile determines the user's permissions, while assigning a role places the user in the role hierarchy. The user configuration should match the responsibilities defined during project planning. Users can then work with the objects and records available to them.

This setup also supports testing because different users can be used to check access behavior. User configuration is therefore a practical step between security design and actual application use. It also provides the basis for validating the project's sharing rules.

PROJECT DETAILS

■ User records connect individual people to the configured Salesforce access model.

■ The project includes a Farm Manager user with the Standard Platform User license, Farm Manager profile and Farm Manager role.

■ The project also includes a Sales Department user associated with a sales-related profile and responsibility.

■ Assigning a profile determines the user's permissions, while assigning a role places the user in the role hierarchy.

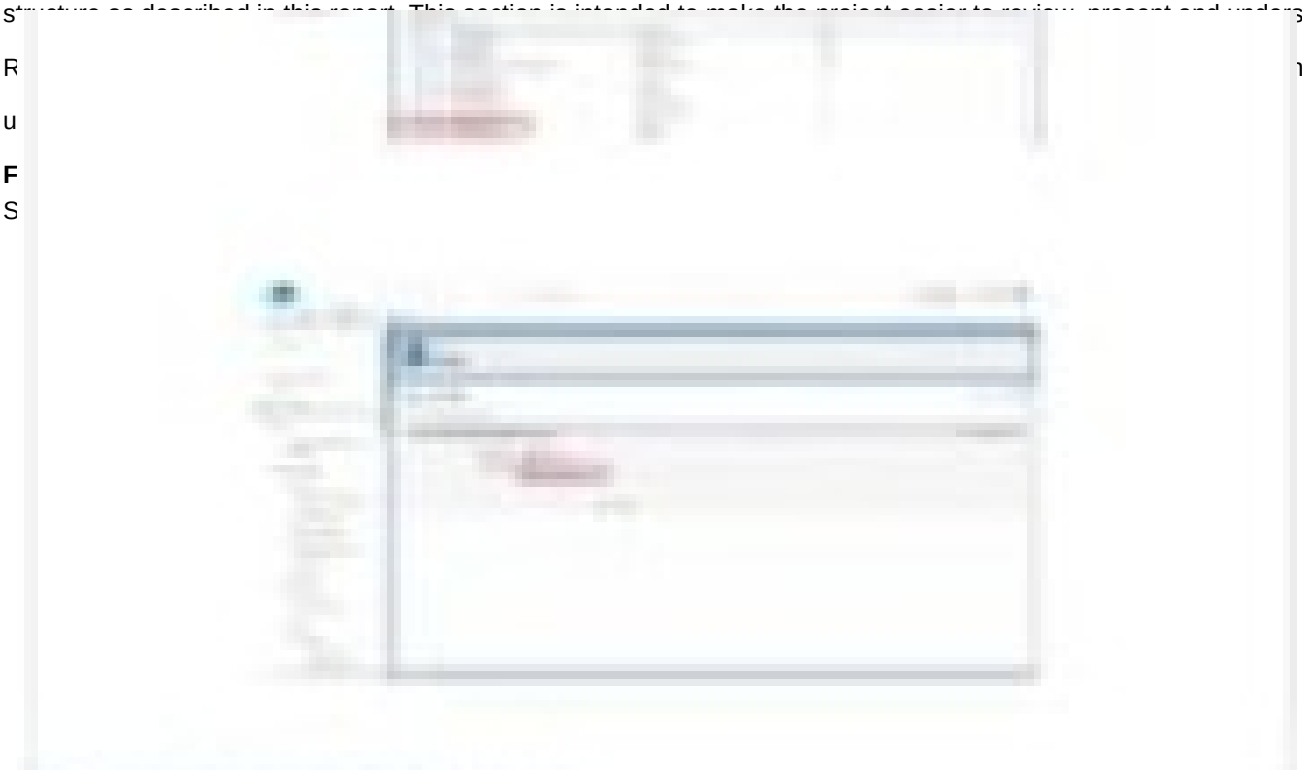
Section relevance: The information in 36. Users forms part of the Farm Management System documentation and is connected to the project's overall data management, Salesforce configuration, security, reporting, testing or future-scope s stand.

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Farm Management System ■ Alternative Version

37. Sharing Rules

Sharing rules are used when additional record access needs to be provided beyond the default configuration. In the project design, Crop records are shared with the Farm Manager using Read/Write access. Buyer records are shared with the Sales

Department using Read/Write access.

These rules support collaboration by making selected records available to the relevant user groups. Sharing rules should be

configured carefully because they affect who can access and modify records. They work together with profiles, roles and other Salesforce security settings.

The purpose of the project sharing configuration is to provide access required for the defined responsibilities. This also creates a security scenario that can be tested using different users. Sharing rules therefore form an important part of the project's access-control design and should be included in the final testing checklist.

PROJECT DETAILS

■ Sharing rules are used when additional record access needs to be provided beyond the default configuration.

■ In the project design, Crop records are shared with the Farm Manager using Read/Write access.

■ Buyer records are shared with the Sales Department using Read/Write access.

■ These rules support collaboration by making selected records available to the relevant user groups.

Section relevance: The information in 37. Sharing Rules forms part of the Farm Management System documentation and is connected to the project's overall data management, Salesforce configuration, security, reporting, testing or future-scope s stand. R n be u **F** S



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38. User Adoption

User adoption depends on how easily users can understand and work with the application. Clear object names, simple navigation and meaningful field labels can reduce confusion. Custom tabs make the main objects easier to reach, while the Lightning App provides a common navigation structure.

Structured picklists can guide users when entering information. Reports and dashboards can make stored information more

useful after records are created. User adoption also depends on appropriate access because users need the permissions required for their responsibilities.

Basic training can help users understand how to create, update and review records. Testing with realistic tasks can identify areas where users may need additional guidance. The project therefore treats user adoption as an important part of successful implementation rather than only a technical configuration activity.

PROJECT DETAILS

■ User adoption depends on how easily users can understand and work with the application.

■ Clear object names, simple navigation and meaningful field labels can reduce confusion.

■ Custom tabs make the main objects easier to reach, while the Lightning App provides a common navigation structure.

■ Structured picklists can guide users when entering information.

Section relevance: The information in 38. User Adoption forms part of the Farm Management System documentation and is connected to the project's overall data management, Salesforce configuration, security, reporting, testing or future-scope structure as described in this report. This section is intended to make the project easier to review, present and understand.

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39. Reports

Reports organize Salesforce records into detailed views that users can inspect and analyze. The project includes a Farmer with Farms report that brings farmer and farm information together. Another report uses Crops and Buyers according to the project design.

Reports can be configured with suitable columns, filters and record groupings so information is easier to understand. The purpose of a report is to convert stored records into a useful view for users. Reports can also provide source data for dashboard components.

When users need detailed information, they can open the report and inspect individual records. This makes reports different

from dashboards, which focus more on summarized visual information. Reports are therefore an important analytics component of the Farm Management System and should be tested with expected project records.

PROJECT DETAILS

■ Reports organize Salesforce records into detailed views that users can inspect and analyze.

■ The project includes a Farmer with Farms report that brings farmer and farm information together.

■ Another report uses Crops and Buyers according to the project design.

■ Reports can be configured with suitable columns, filters and record groupings so information is easier to understand.

Section relevance: The information in 39. Reports forms part of the Farm Management System documentation and is connected to the project's overall data management, Salesforce configuration, security, reporting, testing or future-scope structure as described in this report. This section is intended to make the project easier to review, present and understand.

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40. Dashboard

The dashboard provides a visual summary of selected report information. The project design includes a Farmers with Farms report and a Gauge component based on that report. A dashboard can help users review information quickly without reading every individual record.

The dashboard is connected to report data, so the quality of the displayed information depends on the underlying records and report configuration. Users can use the dashboard for a high-level view and then open the related report for detailed inspection. This creates a two-level analytics process: dashboard for summary and report for detail.

The dashboard also demonstrates the use of Salesforce analytics features within the project. It forms the final layer of the data flow from stored records to useful information and supports quick project monitoring.

PROJECT DETAILS

■ The dashboard provides a visual summary of selected report information.

■ The project design includes a Farmers with Farms report and a Gauge component based on that report.

■ A dashboard can help users review information quickly without reading every individual record.

■ The dashboard is connected to report data, so the quality of the displayed information depends on the underlying records

and report configuration.

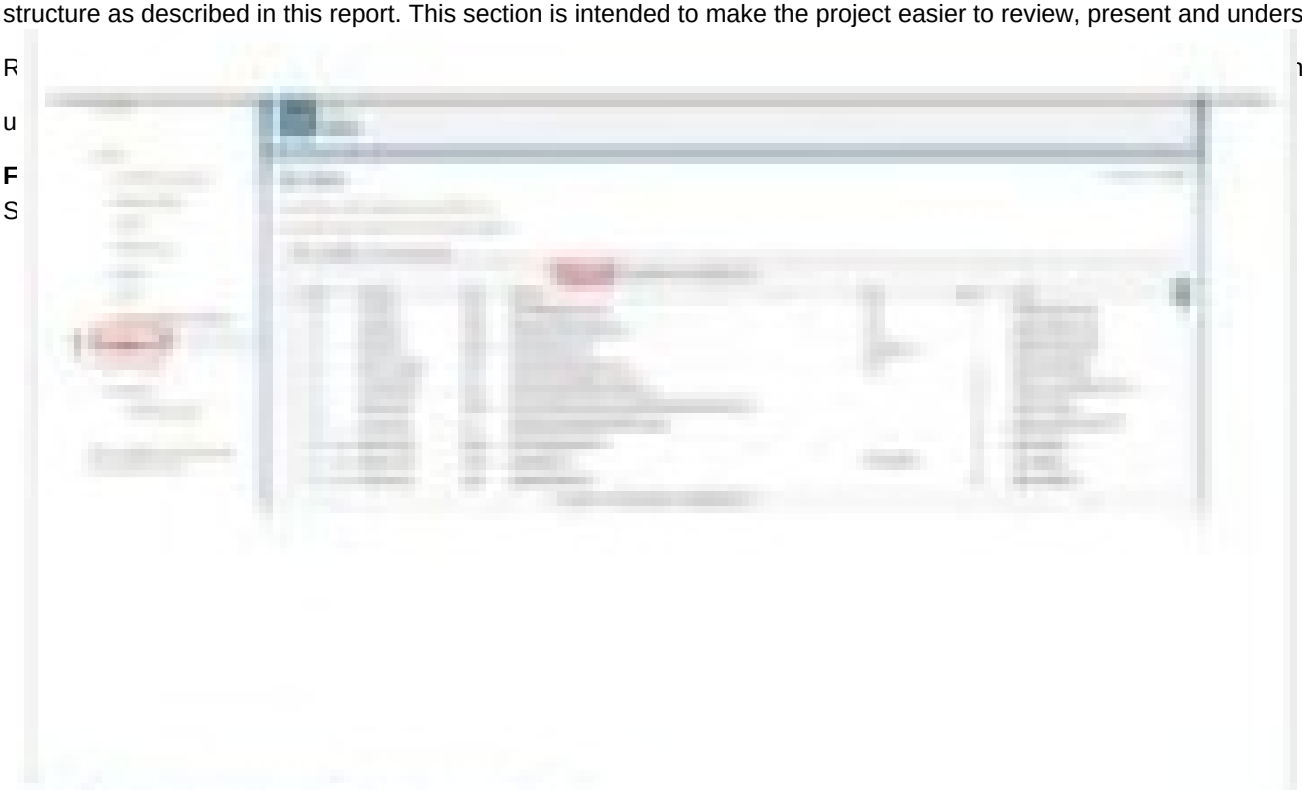
Section relevance: The information in 40. Dashboard forms part of the Farm Management System documentation and is connected to the project's overall data management, Salesforce configuration, security, reporting, testing or future-scope structure as described in this report. This section is intended to make the project easier to review, present and understand.

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41. Analytics Workflow

The analytics workflow begins with records stored in the Farm, Crop, Farmer and Buyer objects. Users first create or update

records so the underlying data is current. Reports then organize selected records into detailed views.

Users can inspect the report to understand individual records and relationships. The dashboard provides a summarized view based on report information. If the dashboard highlights an area that requires attention, the user can open the relevant report and inspect the underlying records.

This workflow connects data entry with analysis instead of treating reports and dashboards as separate features. It also supports monitoring because users can move from summary information to detailed information. The analytics workflow is therefore an important part of the project's overall user experience and shows how stored Salesforce records become useful project information.

PROJECT DETAILS

■ The analytics workflow begins with records stored in the Farm, Crop, Farmer and Buyer objects.

■ Users first create or update records so the underlying data is current.

■ Reports then organize selected records into detailed views.

■ Users can inspect the report to understand individual records and relationships.

Section relevance: The information in 41. Analytics Workflow forms part of the Farm Management System documentation and is connected to the project's overall data management, Salesforce configuration, security, reporting, testing or future-scope structure as described in this report. This section is intended to make the project easier to review, p

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42. Testing Strategy

Testing is performed to check whether the configured Farm Management System behaves according to project requirements. Object testing verifies that Farm, Crop, Farmer and Buyer objects are available. Field testing checks that configured field types and values accept appropriate information.

Relationship testing verifies that Farm can be connected to Farmer and Crop can reference Farm. Security testing checks profiles, roles, users and sharing rules using the relevant user configurations. Report testing checks whether expected records appear in configured reports.

Dashboard testing checks whether the dashboard component displays report information. Testing should also include basic

update and navigation activities. Recording test results provides evidence that the project configuration has been reviewed. This strategy connects implementation work with measurable project verification and provides a basis for final project validation.

PROJECT DETAILS

■ Testing is performed to check whether the configured Farm Management System behaves according to project requirements.

■ Object testing verifies that Farm, Crop, Farmer and Buyer objects are available.

■ Field testing checks that configured field types and values accept appropriate information.

■ Relationship testing verifies that Farm can be connected to Farmer and Crop can reference Farm.

Section relevance: The information in 42. Testing Strategy forms part of the Farm Management System documentation and is connected to the project's overall data management, Salesforce configuration, security, reporting, testing or future-scope structure as described in this report. This section is intended to make the project easier to review, present and understand.

Review note: The project terminology and scope are retained consistently across the documentation. The section can be used as part of the final project explanation and demonstration.

Farm Management System: This page forms part of the same project documentation and supports the overall Salesforce-based management, configuration, security, reporting or testing flow.

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43. Functional Test Cases

Functional test cases focus on expected system behavior. The first test checks whether required custom objects can be opened and records can be created. The second test checks whether configured fields accept intended values and field types.

The third test verifies the relationships between Farmer and Farm and between Farm and Crop. The fourth test checks whether appropriate users can access records assigned to them. The fifth test checks whether sharing rules provide intended Read/Write access.

The sixth test verifies that reports display expected records. The seventh test checks the dashboard component based on the report. These tests can be repeated after configuration changes. Functional testing therefore helps ensure that project features work together rather than only existing individually. It also gives the project a clear set of demonstrations for review

and presentation.

PROJECT DETAILS

■ Functional test cases focus on expected system behavior.

■ The first test checks whether required custom objects can be opened and records can be created.

■ The second test checks whether configured fields accept intended values and field types.

■ The third test verifies the relationships between Farmer and Farm and between Farm and Crop.

Section relevance: The information in 43. Functional Test Cases forms part of the Farm Management System documentation and is connected to the project's overall data management, Salesforce configuration, security, reporting, testing or future-scope structure as described in this report. This section is intended to make the project easier to review, present and understand.

Review note: The project terminology and scope are retained consistently across the documentation. The section can be used as part of the final project explanation and demonstration.

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44. Performance and Reliability

Performance and reliability are considered as part of the project's non-functional requirements. Users should be able to navigate the application and open configured records without unnecessary complexity. Reports and dashboards should load the information provided by their underlying Salesforce configuration.

Reliable record relationships are important because reports and navigation depend on connected data. Consistent field configuration also helps reduce unexpected values that could affect reporting. Testing should include normal record creation, updates, navigation and report access.

The project is implemented within Salesforce, so performance also depends on the configured environment and platform behavior. The goal of this section is to identify the need for predictable operation rather than introduce unrelated performance technology. Reliability supports user confidence in stored information and analytics and should be considered during project testing.

PROJECT DETAILS

■ Performance and reliability are considered as part of the project's non-functional requirements.

■ Users should be able to navigate the application and open configured records without unnecessary complexity.

■ Reports and dashboards should load the information provided by their underlying Salesforce configuration.

■ Reliable record relationships are important because reports and navigation depend on connected data.

Section relevance: The information in 44. Performance and Reliability forms part of the Farm Management System documentation and is connected to the project's overall data management, Salesforce configuration, security, reporting, testing or future-scope structure as described in this report. This section is intended to make the project easier to review, present and understand.

Review note: The project terminology and scope are retained consistently across the documentation. The section can be used as part of the final project explanation and demonstration.

Farm Management System: This page forms part of the same project documentation and supports the overall Salesforce-based management, configuration, security, reporting or testing flow.

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45. Advantages

The Farm Management System provides several practical advantages within the project scope. Centralized records make farm, crop, farmer and buyer information easier to manage in one environment. Structured fields reduce the need for users to enter the same type of information in different formats.

Relationships connect related records and support better navigation. Profiles, roles, users and sharing rules provide a framework for controlled access. Reports provide detailed information, while dashboards provide summarized views.

The system can reduce dependence on manual paperwork and separate files. A Salesforce-based configuration also provides a common platform for collaboration among different user groups. These advantages come directly from features included in the project design. The project therefore combines data management, security and analytics in one application structure and provides a clear basis for organized farm information management.

PROJECT DETAILS

■ The Farm Management System provides several practical advantages within the project scope.

■ Centralized records make farm, crop, farmer and buyer information easier to manage in one environment.

■ Structured fields reduce the need for users to enter the same type of information in different formats.

■ Relationships connect related records and support better navigation.

Section relevance: The information in 45. Advantages forms part of the Farm Management System documentation and is connected to the project's overall data management, Salesforce configuration, security, reporting, testing or future-scope structure as described in this report. This section is intended to make the project easier to review, present and understand.

Review note: The project terminology and scope are retained consistently across the documentation. The section can be used as part of the final project explanation and demonstration.

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46. Limitations

The Farm Management System also has limitations that should be considered. The project requires Salesforce configuration, so users or administrators need familiarity with the platform. Users may require training before they can work confidently with the application.

The quality of reports and dashboards depends on the quality and completeness of records entered into the system. Advanced integrations such as mobile applications, IoT devices or external data services may require additional development. The current project focuses on the defined Farm, Crop, Farmer and Buyer information areas rather than every possible agricultural process.

Security configuration also needs careful maintenance when users or responsibilities change. These limitations do not remove the usefulness of the project, but they identify areas that may need additional work. Understanding limitations helps

define realistic future scope and project boundaries.

PROJECT DETAILS

■ The Farm Management System also has limitations that should be considered.

■ The project requires Salesforce configuration, so users or administrators need familiarity with the platform.

■ Users may require training before they can work confidently with the application.

■ The quality of reports and dashboards depends on the quality and completeness of records entered into the system.

Section relevance: The information in 46. Limitations forms part of the Farm Management System documentation and is connected to the project's overall data management, Salesforce configuration, security, reporting, testing or future-scope structure as described in this report. This section is intended to make the project easier to review, present and understand.

Review note: The project terminology and scope are retained consistently across the documentation. The section can be used as part of the final project explanation and demonstration.

Farm Management System: This page forms part of the same project documentation and supports the overall Salesforce-based management, configuration, security, reporting or testing flow.

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47. Conclusion

The Farm Management System provides a structured Salesforce approach to managing farm-related information. The project uses four main custom objects to represent Farm, Crop, Farmer and Buyer records. Fields provide the required information, while relationships connect related records.

Custom tabs and a Lightning App provide navigation for users. Profiles, roles, users and sharing rules provide the security and access structure. Reports and dashboards convert stored information into useful analytical views.

Testing covers the major configuration areas and helps verify expected behavior. The project also considers user adoption so the system is understandable and practical. Overall, the project demonstrates how Salesforce features can be combined

to create a centralized agricultural information management solution. The conclusion connects the original problem with the configured solution and future development possibilities.

PROJECT DETAILS

■ The Farm Management System provides a structured Salesforce approach to managing farm-related information.

■ The project uses four main custom objects to represent Farm, Crop, Farmer and Buyer records.

■ Fields provide the required information, while relationships connect related records.

■ Custom tabs and a Lightning App provide navigation for users.

Section relevance: The information in 47. Conclusion forms part of the Farm Management System documentation and is connected to the project's overall data management, Salesforce configuration, security, reporting, testing or future-scope structure as described in this report. This section is intended to make the project easier to review, present and understand.

Review note: The project terminology and scope are retained consistently across the documentation. The section can be used as part of the final project explanation and demonstration.

Farm Management System: This page forms part of the same project documentation and supports the overall Salesforce-based management, configuration, security, reporting or testing flow.

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48. Future Scope

The project identifies several possible future enhancements. Mobile application integration could make farm information accessible through a mobile workflow. AI-based crop prediction could be explored as an extension for analyzing crop-related information.

Weather data integration could provide additional environmental information alongside farm and crop records. IoT sensor connectivity could allow suitable field information to be connected with the management system. Government scheme integration could help organize information related to relevant agricultural schemes.

Market price forecasting is another future possibility that could support additional analysis. These future features are listed as extensions rather than current project functions. Each enhancement would require additional design, integration and testing. The future scope demonstrates that the Salesforce-based foundation can provide a starting point for broader agricultural information management.

PROJECT DETAILS

■ The project identifies several possible future enhancements.

■ Mobile application integration could make farm information accessible through a mobile workflow.

■ AI-based crop prediction could be explored as an extension for analyzing crop-related information.

■ Weather data integration could provide additional environmental information alongside farm and crop records.

Section relevance: The information in 48. Future Scope forms part of the Farm Management System documentation and is connected to the project's overall data management, Salesforce configuration, security, reporting, testing or future-scope structure as described in this report. This section is intended to make the project easier to review, present and understand.

Review note: The project terminology and scope are retained consistently across the documentation. The section can be used as part of the final project explanation and demonstration.

Farm Management System: This page forms part of the same project documentation and supports the overall Salesforce-based management, configuration, security, reporting or testing flow.

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49. Appendix – Object Checklist

The project uses four core custom objects: Farm, Crop, Farmer and Buyer. Farm stores information about the farm and is connected to Farmer through the configured Master-Detail relationship. Crop stores production information and uses a Lookup relationship to Farm.

Farmer stores farmer contact and location information. Buyer stores buyer contact, address and buyer type information. Each object is configured to support reports and search according to the project design.

The objects form the foundation of the application data model. Custom tabs provide navigation to these records. Reports and dashboards use information from these objects for analysis. The object checklist can be used during project review to confirm that all four main areas have been configured and are available for testing and demonstration.

PROJECT DETAILS

■ The project uses four core custom objects: Farm, Crop, Farmer and Buyer.

■ Farm stores information about the farm and is connected to Farmer through the configured Master-Detail relationship.

■ Crop stores production information and uses a Lookup relationship to Farm.

■ Farmer stores farmer contact and location information.

Section relevance: The information in 49. Appendix – Object Checklist forms part of the Farm Management System documentation and is connected to the project's overall data management, Salesforce configuration, security, reporting, testing or future-scope structure as described in this report. This section is intended to make the project easier to review, present and understand.

Review note: The project terminology and scope are retained consistently across the documentation. The section can be used as part of the final project explanation and demonstration.

Farm Management System: This page forms part of the same project documentation and supports the overall Salesforce-based management, configuration, security, reporting or testing flow.

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50. Appendix – Field Checklist

The Farm field checklist includes Farm Type, Soil Quality, Irrigation Type, Location, Size and the Farmer relationship. The Crop field checklist includes Crop Name, Planting Date, Harvest Date, Farm lookup, Actual Yield, Total Acres Harvested and Expected Yield per Acre formula. The Farmer field checklist includes Contact Number, State, District and Village. The Buyer field checklist includes Contact, Email, Address and Buyer Type. These fields represent the main information described in the project design. Field types are selected according to the kind of information being stored.

Picklists provide controlled choices where required. Relationships connect records across objects. The checklist can be used to review whether the configured data model contains the required fields before testing, reporting and final project demonstration.

PROJECT DETAILS

■ The Farm field checklist includes Farm Type, Soil Quality, Irrigation Type, Location, Size and the Farmer relationship.

■ The Crop field checklist includes Crop Name, Planting Date, Harvest Date, Farm lookup, Actual Yield, Total Acres Harvested

and Expected Yield per Acre formula.

■ The Farmer field checklist includes Contact Number, State, District and Village.

■ The Buyer field checklist includes Contact, Email, Address and Buyer Type.

Section relevance: The information in 50. Appendix – Field Checklist forms part of the Farm Management System documentation and is connected to the project's overall data management, Salesforce configuration, security, reporting, testing or future-scope structure as described in this report. This section is intended to make the project easier to review, present and understand.

Review note: The project terminology and scope are retained consistently across the documentation. The section can be used as part of the final project explanation and demonstration.

Farm Management System: This page forms part of the same project documentation and supports the overall Salesforce-based management, configuration, security, reporting or testing flow.

Farm Management System ■ Alternative Version

51. Appendix – Security Checklist

The security checklist includes profiles, role hierarchy, users and sharing rules. The project design clones the Standard User profile to create role-specific profiles for Farm Manager and Sales Department responsibilities. The Farm Manager role is positioned under CEO in the role hierarchy.

Users are assigned suitable profiles and roles. Sharing rules provide additional access where required by the project. Crop records are shared with Farm Manager using Read/Write access, while Buyer records are shared with Sales Department using Read/Write access.

These components should be reviewed together because access behavior depends on their combined configuration. Security testing should use the relevant user records to confirm intended access. The checklist therefore provides a structured way to review the project's access-control setup before final submission.

PROJECT DETAILS

■ The security checklist includes profiles, role hierarchy, users and sharing rules.

■ The project design clones the Standard User profile to create role-specific profiles for Farm Manager and Sales Department

responsibilities.

■ The Farm Manager role is positioned under CEO in the role hierarchy.

■ Users are assigned suitable profiles and roles.

Section relevance: The information in 51. Appendix – Security Checklist forms part of the Farm Management System documentation and is connected to the project's overall data management, Salesforce configuration, security, reporting, testing or future-scope structure as described in this report. This section is intended to make the project easier to review, present and understand.

Review note: The project terminology and scope are retained consistently across the documentation. The section can be used as part of the final project explanation and demonstration.

55. FINAL REVIEW, APPENDIX & PROJECT SUMMARY

52. Appendix – Analytics Checklist

The analytics checklist includes reports and dashboards. The project includes a Farmer with Farms report that combines farmer and

farm information. Another report uses Crops and Buyers according to the project design.

Reports should be checked to ensure expected records appear with suitable columns and filters. The dashboard includes a Gauge

component based on the Farmers with Farms report. Dashboard testing should verify that report data is available and displayed

correctly.

KEY REVIEW POINTS

■ The analytics checklist includes reports and dashboards.

■ The project includes a Farmer with Farms report that combines farmer and farm information.

■ Another report uses Crops and Buyers according to the project design.

■ Reports should be checked to ensure expected records appear with suitable columns and filters.

53. Appendix – Implementation Checklist

The implementation checklist provides a complete sequence for reviewing the project. First, confirm that the four custom objects

have been created. Next, confirm that custom tabs and the Lightning App provide the required navigation.

Review the fields and relationships for each object. Confirm the profiles and role hierarchy, then review the user configurations.

Check the sharing rules and verify the intended Read/Write access.

KEY REVIEW POINTS

■ The implementation checklist provides a complete sequence for reviewing the project.

■ First, confirm that the four custom objects have been created.

Farm Management System: This page forms part of the same project documentation and supports the overall Salesforce-based management, configuration, security, reporting or testing flow.